SLIDE 1 – Title

Separation and Purification of Human Serum Albumin from Blood Serum Using Surface-Modified Bentonite

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SLIDE 2 – Scientific Context: Albumin in Plasma

Composition of Blood Plasma

[Insert plasma composition figure]

Human Serum Albumin (HSA) is the most abundant protein in blood plasma.

- 585 amino acids
- ~69 kDa molecular weight
- Synthesized in liver (~12 g/day)
- Responsible for 75–80% of plasma oncotic pressure

Due to its high concentration and multifunctionality, albumin is a major target in therapeutic purification.

SLIDE 3 – Biological & Clinical Importance

Albumin plays a central physiological role:

- Transport of fatty acids, hormones, and drugs
- Osmotic pressure regulation
- Plasma buffering

Clinical applications:

- Drug delivery systems
- Plasma expanders
- Biopharmaceutical formulations

The increasing medical demand requires efficient and scalable purification strategies.

SLIDE 4 – The Core Separation Problem

Serum contains multiple proteins with:

• Overlapping molecular weight
• Similar surface charge at physiological pH
• Competitive adsorption behavior

Therefore, selective separation of albumin from complex biological matrices remains challenging.

SLIDE 5 – Why Albumin Is Difficult to Separate

• Globular structure similar to IgG
• Comparable physicochemical properties
• Overlapping isoelectric regions
• Strong intermolecular interactions in serum

These similarities complicate conventional separation.

SLIDE 6 – Limitations of Current Technologies

Conventional purification relies on chromatography:

• Ion-exchange
• Affinity chromatography
• Size exclusion

Although effective, they suffer from:

• High resin cost
• Multi-step operation
• Energy-intensive processing
• Limited reusability

Alternative low-cost approaches are required.

SLIDE 7 – Patent & Market Perspective

[Insert albumin market growth chart]
[Insert patent statistics figure]

Patent trends and market expansion show strong industrial interest.

However, cost-effective adsorption platforms remain underexplored.

SLIDE 8 – Identified Research Gap

There is no method that simultaneously provides:

- High selectivity
- Low cost
- Regenerability
- Mechanistic clarity
- Real-serum applicability

This gap motivated the present work.

SLIDE 9 – Conceptual Framework

We proposed a rational adsorption-based strategy:

- Natural bentonite as base material
- Surface modification to tailor affinity
- Integration of molecular modeling and experimental validation

This approach enables controlled surface–protein interaction design.

SLIDE 10 – Why Bentonite?

Bentonite offers:

- Abundance and low cost
- High surface area layered structure
- Tunable surface chemistry
- Chemical and thermal stability

However, raw bentonite lacks selectivity.

SLIDE 11 – Need for Surface Functionalization

To achieve selective albumin adsorption:

• Electrostatic interactions must be engineered
• Hydrophobic interactions must be balanced
• Surface charge must be tailored

Thus, surface modification becomes essential.

SLIDE 12 – Molecular Docking-Based Precursor Screening

Several silane precursors (as initially considered in the design stage) were computationally evaluated.

Molecular docking simulations were performed to:

• Compare binding energy with HSA
• Analyze interaction modes
• Predict selectivity potential

Among the screened candidates, **DTSACL exhibited the most favorable predicted interaction with HSA**.

Therefore, DTSACL was selected as the surface modifier.

[Insert comparative docking energy graph of silane candidates]

SLIDE 13 – DTSACL Functionalization Strategy

DTSACL contains:

• Quaternary ammonium headgroup → electrostatic attraction
• C18 alkyl chain → hydrophobic enhancement
• Silane functionality → covalent anchoring

This design allows balanced affinity rather than irreversible binding.

[Insert chemical structure]

SLIDE 14 – Docking Analysis for Selectivity (HSA vs IgG)

Docking simulations were performed for:

• Human Serum Albumin (HSA)
• Immunoglobulin G (IgG)

Objective:

To evaluate selectivity of DTSACL-modified surface toward major serum proteins.

Results showed:

• Lower binding energy for HSA
• More favorable binding orientation
• Weaker and less optimal interaction with IgG

This computational comparison predicted selective adsorption behavior.

[Insert HSA vs IgG docking comparison graph]

SLIDE 15 – Experimental Workflow

Two integrated phases:

1. Computational screening and selectivity prediction

2. Laboratory synthesis and validation

This combined approach strengthens mechanistic reliability.

SLIDE 16 – Synthesis of DTSACL-Modified Bentonite

• Bentonite pre-dried at 105°C
• Reaction with 20 wt% DTSACL under reflux
• Heating from 70 to 120°C
• 2-hour reaction
• Ethanol washing
• Final drying

This ensured stable covalent grafting.

SLIDE 17 – Material Characterization

[Insert FTIR spectrum]
[Insert XRD pattern]
[Insert TGA curve]
[Insert SEM image]

FTIR confirmed functional groups.
XRD indicated structural integrity.
TGA demonstrated organic loading.
SEM revealed surface morphology changes.

SLIDE 18 – Experimental Design (CCD)

Three variables:

- pH
- Initial HSA concentration
- Adsorbent dosage

Central Composite Design allowed:

- Statistical optimization
- Interaction analysis
- Response surface modeling

[Insert ANOVA and response surface plots]

SLIDE 19 – Adsorption Experiments

Batch adsorption under controlled conditions.

- pH adjustment
- Controlled agitation
- Centrifugation
- HPLC quantification

[Insert HPLC chromatogram]

SLIDE 20 – Isotherm Analysis

Langmuir and Freundlich models were applied.

Results suggested:

- Monolayer-dominated adsorption
- Surface heterogeneity

[Insert isotherm plots]

SLIDE 21 – Kinetic Study

Pseudo-second-order model provided best fit.

Indicates chemisorption-dominated interaction mechanism.

[Insert kinetic plots]

SLIDE 22 – Thermodynamic Analysis

$\Delta G° < 0$ → spontaneous
$\Delta H°$ → interaction nature
$\Delta S°$ → interfacial entropy change

[Insert thermodynamic graphs]

Adsorption process is thermodynamically favorable.

SLIDE 23 – Regeneration & Recyclability

Three regeneration systems tested.

Ethanol-based elution proved effective.

Four adsorption–desorption cycles showed stable performance.

[Insert SDS-PAGE gel image]

SLIDE 24 – Selectivity Validation (Experimental)

IgG adsorption evaluated under optimized HSA conditions.

Results confirmed:

• Lower IgG adsorption capacity
• Different kinetic behavior

Consistent with molecular docking predictions.

[Insert comparative adsorption graph]

SLIDE 25 – Effect of pH and Surface Charge

[Insert pH adsorption curve]
[Insert zeta potential plot if available]

Maximum adsorption observed around pH 6.2.

This aligns with electrostatic interaction between modified surface and albumin charge distribution.

SLIDE 26 – Packed Column Experiment

Continuous flow system implemented.

Breakthrough behavior monitored.

[Insert breakthrough curve]

Demonstrates applicability in dynamic conditions.

SLIDE 27 – Binary Protein Separation

HSA/IgG mixture separation performed.

Selective retention observed.

[Insert binary chromatogram]

SLIDE 28 – Real Human Serum Application

Real serum applied to packed column.

Eluted fractions analyzed by SDS-PAGE.

[Insert gel image]

Demonstrates real-matrix performance.

SLIDE 29 – Overall Performance Summary

✓ Selective adsorption
✓ Regenerability
✓ Reproducibility
✓ Real serum validation
✓ Mechanistic support via docking

SLIDE 30 – Scientific Contribution

1. Rational silane precursor screening
2. Computational prediction of selectivity
3. Mechanistic adsorption modeling
4. Practical demonstration in real serum

SLIDE 31 – Industrial Outlook

Potential advantages over chromatography:

- Lower cost
- Reusable adsorbent
- Scalable packed column design
- Simplified purification pathway

SLIDE 32 – Final Conclusion

DTSACL-modified bentonite provides:

- Controlled surface affinity
- Selective HSA adsorption over IgG
- Reusability and stability
- Applicability to real biological systems

This work presents a promising adsorption-based alternative for protein purification